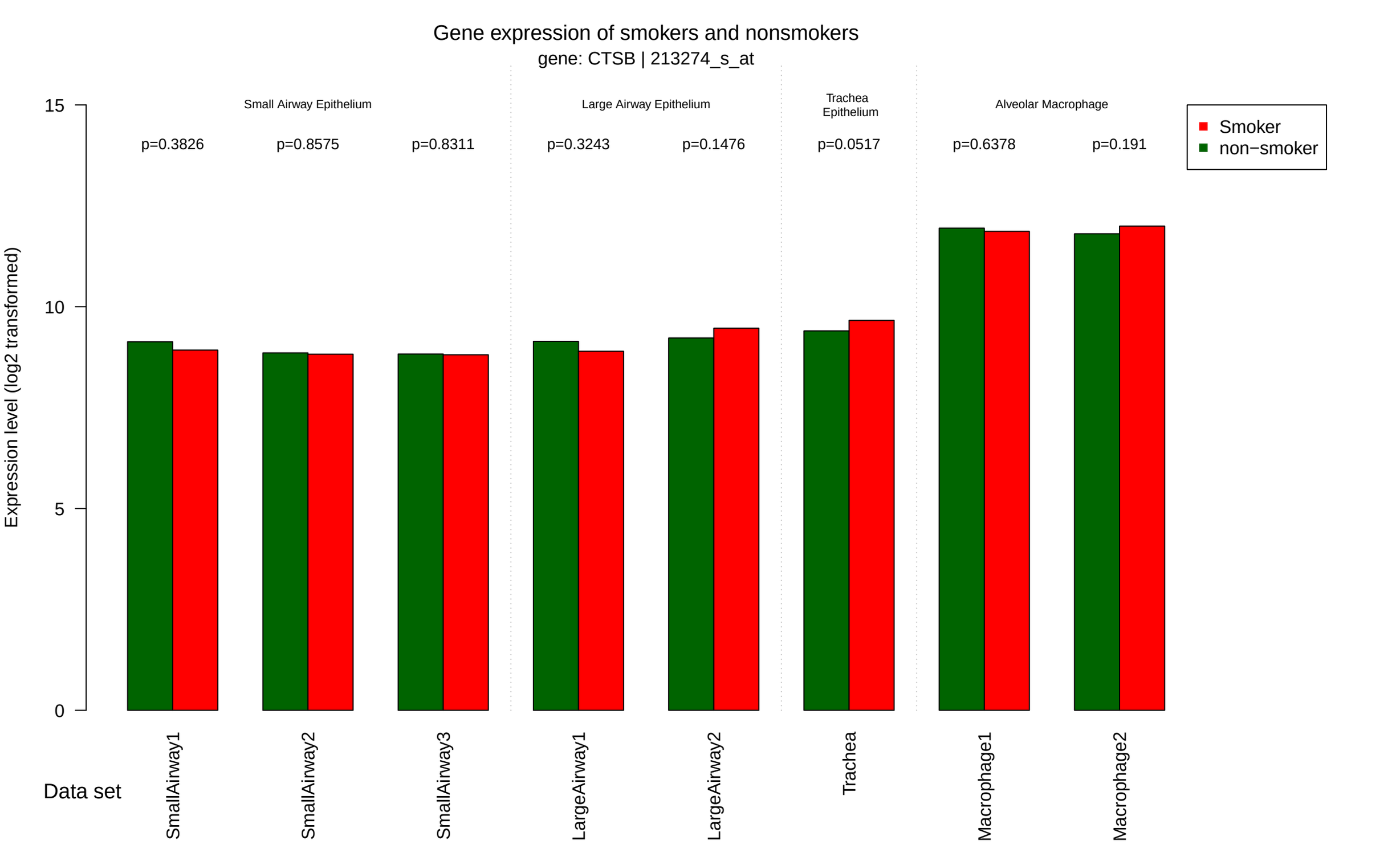
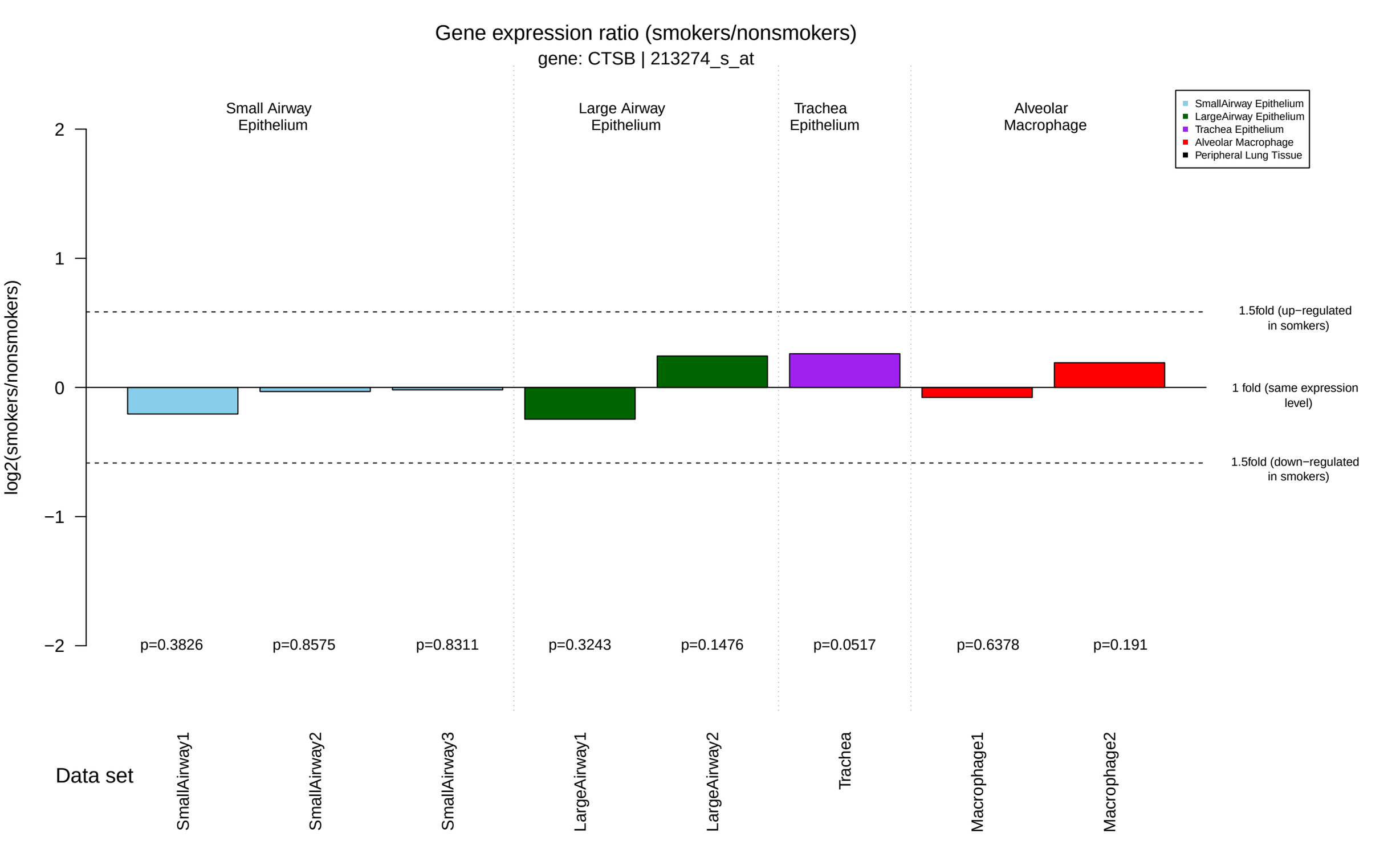
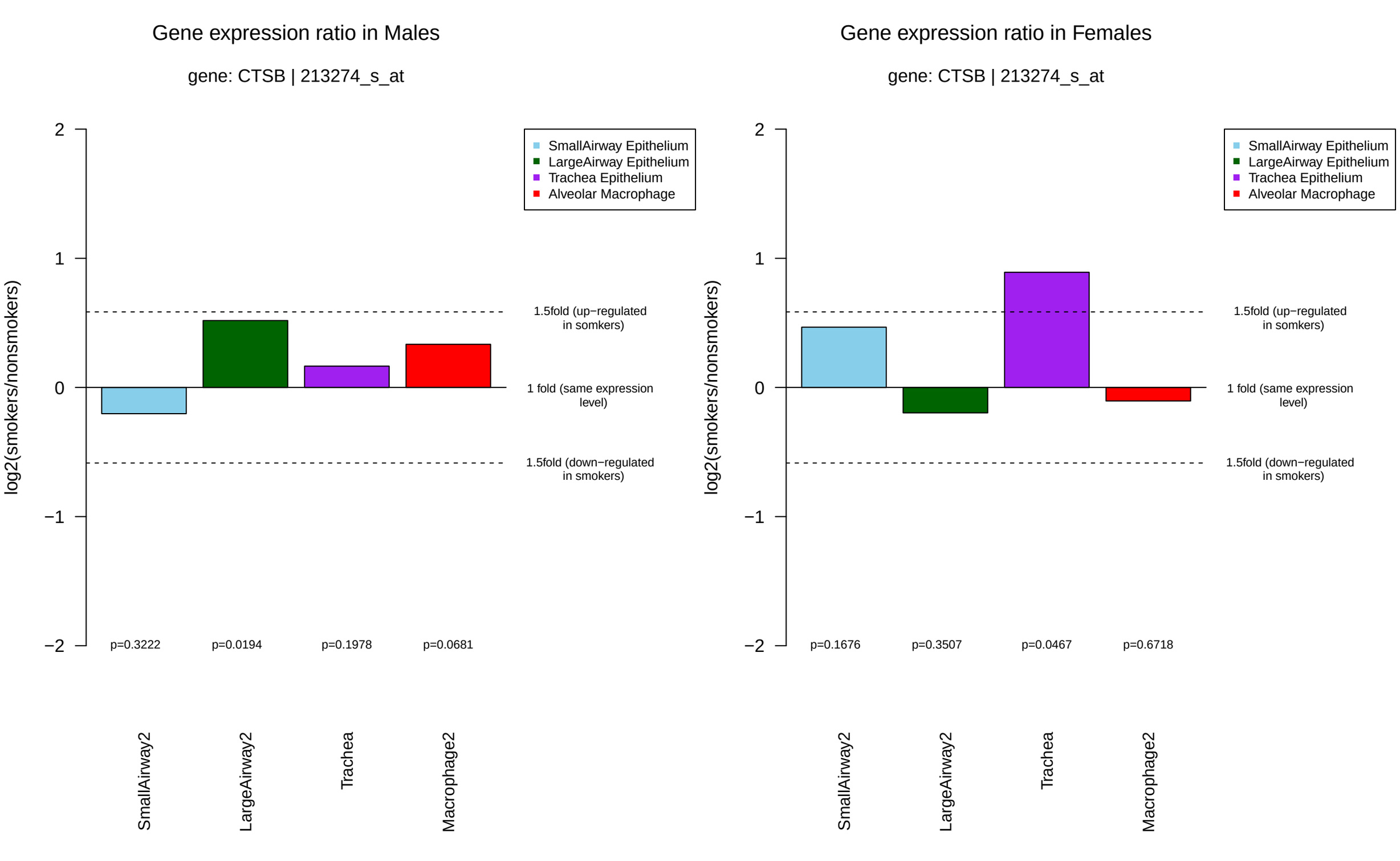
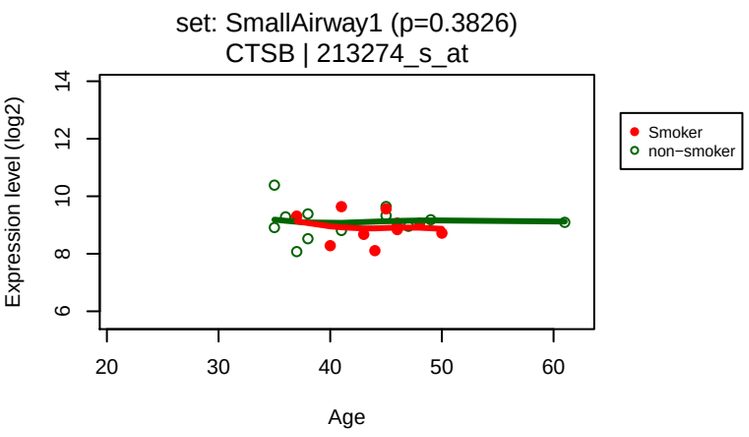
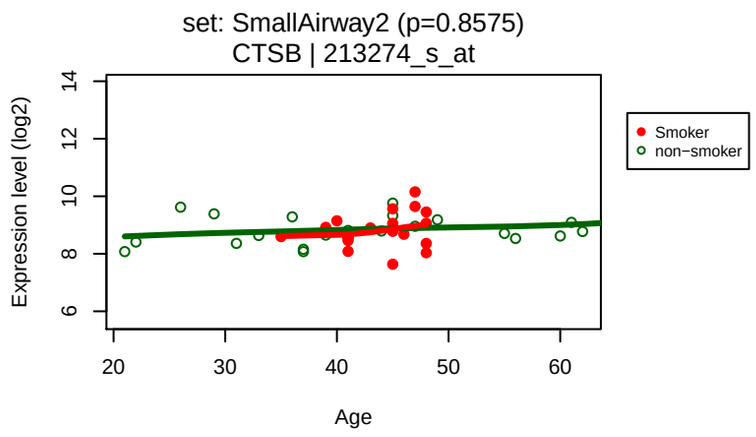
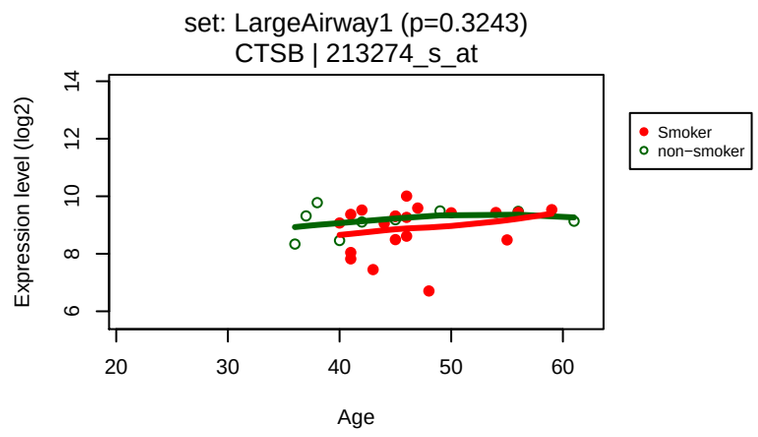

Gene expression of smokers and nonsmokers
gene: CTSB | 213274_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3826
p=0.8575
p=0.8311
p=0.3243
p=0.1476
p=0.0517
p=0.6378
p=0.191
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

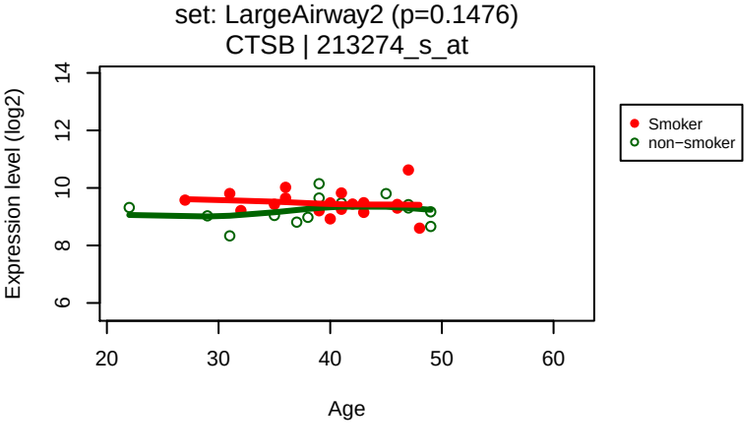

Gene expression ratio (smokers/nonsmokers)
gene: CTSB | 213274_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3826
p=0.8575
p=0.8311
p=0.3243
p=0.1476
p=0.0517
p=0.6378
p=0.191
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

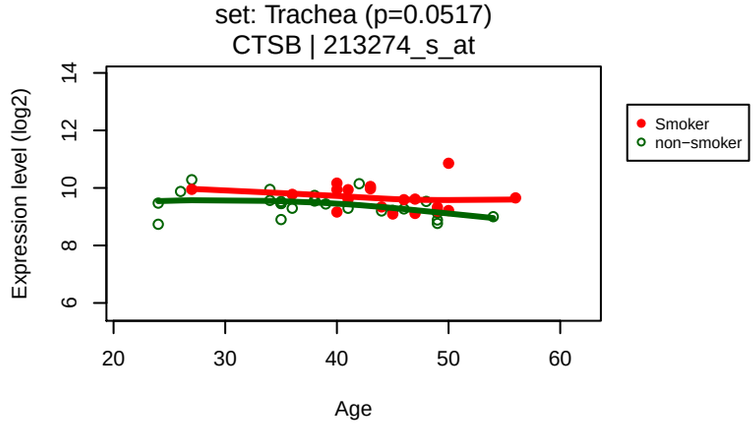

Gene expression ratio in Males
gene: CTSB | 213274_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3222
p=0.0194
p=0.1978
p=0.0681
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: CTSB | 213274_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1676
p=0.3507
p=0.0467
p=0.6718
SmallAirway2
LargeAirway2
Trachea
Macrophage2

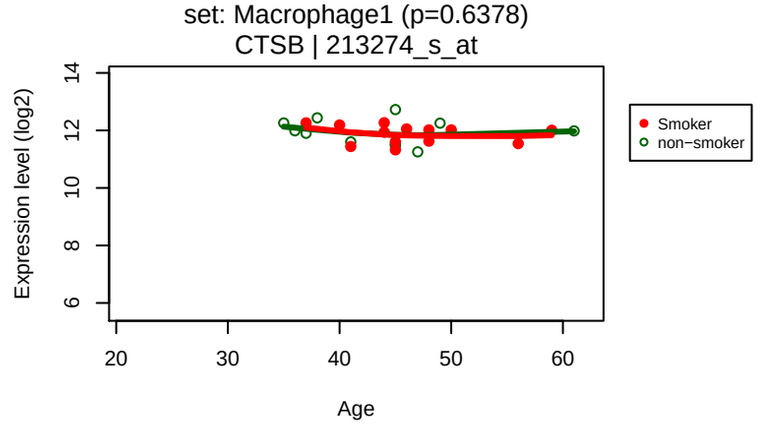

set: SmallAirway1 (p=0.3826)
CTSB | 213274_s_at
Expression level (log2)
Age
Smoker
non−smoker

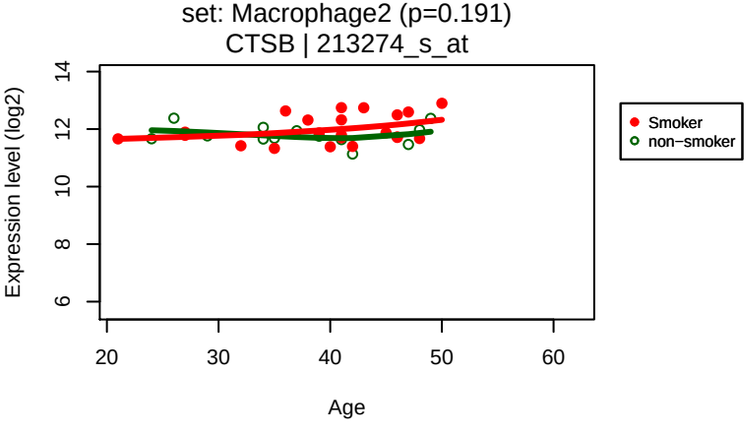

set: SmallAirway2 (p=0.8575)
CTSB | 213274_s_at
Expression level (log2)
Age
Smoker
non−smoker

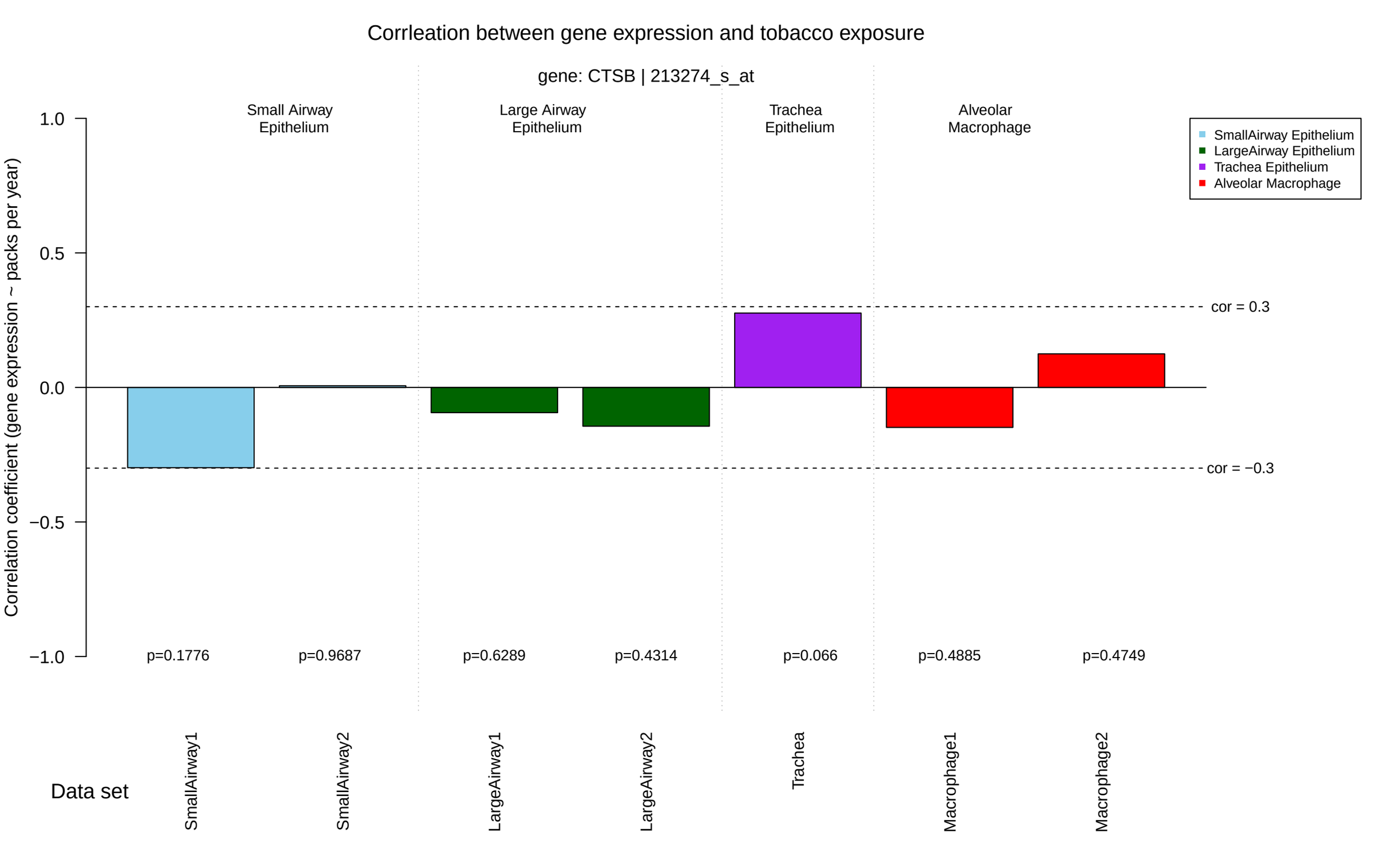

set: LargeAirway1 (p=0.3243)
CTSB | 213274_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1476)
CTSB | 213274_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0517)
CTSB | 213274_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6378)
CTSB | 213274_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.191)
CTSB | 213274_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: CTSB | 213274_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1776
p=0.9687
p=0.6289
p=0.4314
p=0.066
p=0.4885
p=0.4749
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2